FUEL PUMP CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

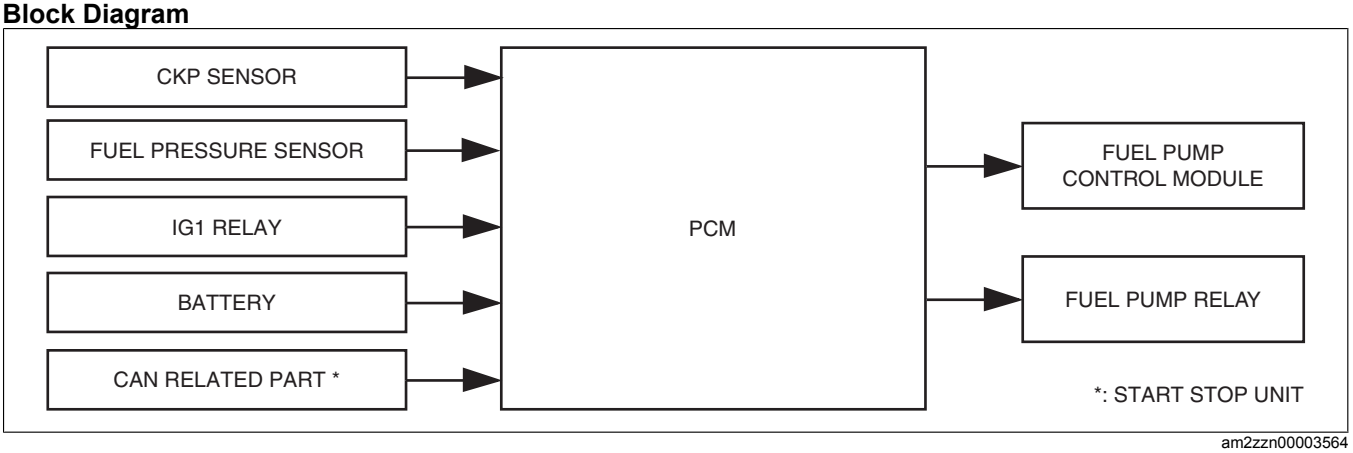
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With Fuel Pump Control Module

Outline

- By switching the fuel pump discharge amount, reduced power consumption and improved fuel economy have been achieved.
- The PCM determines the optimum fuel pump discharge amount according to the engine operation conditions, and sends the fuel pump drive signal to the fuel pump control module.

Block Diagram



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Operation

Fuel pump relay

- The PCM switches the fuel pump relay on/off according to the engine operation conditions.

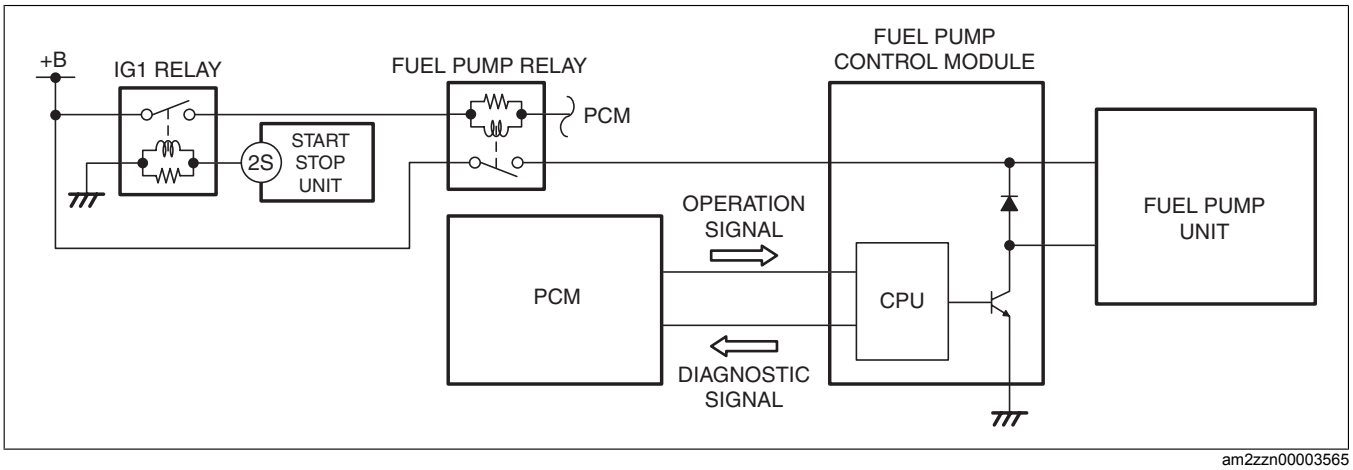
Relay ON/OFF	Control conditions
ON	• When any of the following conditions is met: — When ignition is switched ON (**for approx. 1 s**) — During engine cranking — While engine is running — When engine is stopped by i-stop control
OFF	• When engine is stopped (excludes engine stop by i-stop control) • Immobilizer system related information (engine start prohibited) received from start stop unit • Collision signal from SAS control module is received

Fuel pump control module

- The PCM determines the fuel pump discharge amount according to the engine operation conditions, and controls the output duty ratio to the fuel pump control module according to the discharge amount.

Output duty ratio	Control conditions
95%	• When any of the following conditions is met: — At engine start — Fuel pump control module malfunction (if request voltage is unobtainable)
10—90%	• Except for control conditions with output duty ratio of 5% or 95%
5%	• When engine is stopped (excludes engine stop by i-stop control)

Fuel pump operation circuit diagram



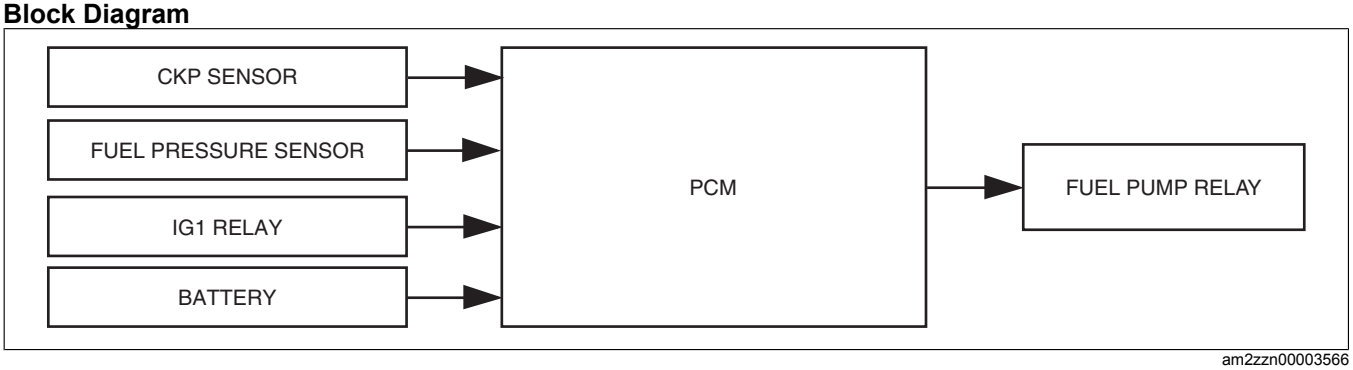
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Without Fuel Pump Control Module

Outline

- Performs energization of fuel pump relay only when engine is running (operates the fuel pump) to improve safety and durability.

Block Diagram



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Operation

Operation condition

- When the ignition is switched START, the PCM turns the fuel pump relay on for **1 s**, then off.
- When it is detected that the CKP signal falls during cranking, the fuel pump relay turns on.
- When the engine is stopped, the fuel pump relay turns off.